

DECK INDEX • PART 1

01:

Newton's 2nd Law

02:

Gravitation

03:

Velocity-Time

04:

05:

06:

07:

08:

09:

10:

11:

12:

13:

14:

15:

YAADCARD.IN

NEWTON'S
2ND
LAW

YAADCARD.IN

GRAVITATION

YAADCARD.IN

VELOCITY-TIME

YAADCARD.IN

DISPLACEMENT

YAADCARD.IN

VELOCITY-SQUARED

YAADCARD.IN

WORK
DONE

YAADCARD.IN

KINETIC
ENERGY

YAADCARD.IN

POTENTIAL
ENERGY

YAADCARD.IN

POWER

YAADCARD.IN

MOMENTUM

YAADCARD.IN

IMPULSE

YAADCARD.IN

FLASH CARDS

PHYSICS FORMULAS

yaadcard.in

YAADCARD • PHYSICS FORMULAS

PRODUCT DESCRIPTION

Premium high-yield study deck engineered for quick revision and active recall. Contains concise cards covering major core concepts, visual diagrams, mathematical formulas, and memory mnemonics. Perfect for exam preparation, competitive tests, and classroom revision.

Pack Contents:

Premium Flashcards & reference insert

Subject:

Physics Formulas

MRP:

₹250.00 (incl. of all taxes)

Support:

support@yaadcard.in

YAADCARD Publications
Made in India • Recyclable

▼ FOLD TOP ▼

▼ FOLD BOTTOM ▼

VELOCITY-SQUARED

Third kinematic equation

Relates velocity squared to acceleration and displacement without needing time.

Formula: $v^2 = u^2 + 2as$

Useful when time is unknown

DISPLACEMENT

Second kinematic equation

Calculates distance traveled given initial velocity, time, and constant acceleration.

Formula: $s = ut + \frac{1}{2}at^2$

s is total displacement

VELOCITY-TIME

First kinematic equation

Relates final velocity to initial velocity, acceleration, and time elapsed.

Formula: $v = u + at$

u=initial, v=final velocity

GRAVITATION

Newton's Law of Gravity

Every mass attracts every other mass with a force proportional to their masses.

Formula: $F = \frac{Gm_1m_2}{r^2}$

G is universal gravitational constant

NEWTON'S 2ND LAW

Force equals mass times acceleration

Relates the force applied to an object to its mass and resulting acceleration.

Formula: $F = ma$

Foundation of classical mechanics

DECK INDEX • PART 2

16: Wave Equation

17: Snell's Law

18: Lens Formula

19: Mass-Energy

20: Photon Energy

21: de Broglie

22: Uncertainty Principle

23: Boyle's Law

24: Ideal Gas Law

25: Bernoulli's Equation

26: Escape Velocity

27: Centripetal Force

28: Faraday's Law

29: Doppler Effect

YAADCARD.IN

POTENTIAL ENERGY

Stored gravitational energy

Energy an object possesses due to its position in a gravitational field.

Formula: $PE = mgh$

h is height above reference

KINETIC ENERGY

Energy of motion

The energy possessed by an object due to its motion, dependent on mass and velocity.

Formula: $KE = \frac{1}{2}mv^2$

Scalar quantity, unit Joule

WORK DONE

Energy transferred by force

Work equals the force applied times displacement times the cosine of the angle.

Formula: $W = Fd \cos \theta$

SI unit: Joule (J)

IMPULSE

Force applied over time

Equals the change in momentum caused by a force acting over a time interval.

Formula: $J = F\Delta t = \Delta p$

Same units as momentum

MOMENTUM

Mass in motion quantity

The product of an object's mass and velocity; conserved in closed systems.

Formula: $p = mv$

Conserved in collisions

POWER

Rate of doing work

The rate at which work is done or energy is transferred over time.

Formula: $P = \frac{W}{t}$

SI unit: Watt (W)

OHM'S
LAW

YAADCARD.IN

ELECTRICAL
POWER

YAADCARD.IN

COULOMB'S
LAW

YAADCARD.IN

CAPACITANCE

YAADCARD.IN

WAVE
EQUATION

YAADCARD.IN

SNELL'S
LAW

YAADCARD.IN

LENS
FORMULA

YAADCARD.IN

MASS-ENERGY

YAADCARD.IN

PHOTON
ENERGY

YAADCARD.IN

DE
BROGLIE

YAADCARD.IN

UNCERTAINTY
PRINCIPLE

YAADCARD.IN

BOYLE'S
LAW

YAADCARD.IN

IDEAL
GAS
LAW

YAADCARD.IN

BERNOULLI'S
EQUATION

YAADCARD.IN

ESCAPE
VELOCITY

YAADCARD.IN

CENTRIPETAL
FORCE

YAADCARD.IN

FARADAY'S
LAW

YAADCARD.IN

DOPPLER
EFFECT

YAADCARD.IN

SNELL'S LAW Light bending at surfaces Describes how light refracts when passing between two media of different densities. Formula: $n_1 \sin \theta_1 = n_2 \sin \theta_2$ Governs lens & prism refraction	WAVE EQUATION Speed, frequency, wavelength link Wave speed equals the product of frequency and wavelength for any wave. Formula: $v = f\lambda$ Applies to all wave types	CAPACITANCE Charge storage capacity Ratio of stored charge to voltage; energy storage depends on capacitance and voltage. Formula: $C = Q/V$ Energy: $E = \frac{1}{2}CV^2$	COULOMB'S LAW Force between charges The electrostatic force between two point charges is proportional to their product. Formula: $F = \frac{kq_1q_2}{r^2}$ k is Coulomb's constant	ELECTRICAL POWER Energy consumption rate Rate of electrical energy transfer, expressible via voltage, current, or resistance. Formula: $P = VI = I^2R$ Also: $P = V^2/R$	OHM'S LAW Voltage, current, resistance link Voltage across a conductor is directly proportional to the current through it. Formula: $V = IR$ Georg Ohm, foundational law
BOYLE'S LAW Pressure-volume relationship At constant temperature, pressure and volume of a gas are inversely proportional. Formula: $PV = \text{constant}$ Valid only at constant T	UNCERTAINTY Limits of quantum precision Position and momentum of a particle cannot both be precisely known simultaneously. Formula: $\Delta x \Delta p \geq \hbar/4\pi$ Werner Heisenberg, 1927	DE BROGLIE Matter has wave nature Every moving particle has an associated wavelength inversely proportional to momentum. Formula: $\lambda = \hbar/p$ Basis of wave-particle duality	PHOTON ENERGY Energy of a light quantum The energy of a photon is proportional to its frequency, via Planck's constant. Formula: $E = \hbar f$ Max Planck, quantum theory origin	MASS-ENERGY Einstein's famous equation Mass and energy are equivalent and interconvertible, with light-speed as the constant. Formula: $E = mc^2$ Proposed by Albert Einstein	LENS FORMULA Relates image and object distance Connects focal length to object and image distances for a thin lens. Formula: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ Sign convention critical here
DOPPLER EFFECT Frequency shift from motion Apparent change in wave frequency due to relative motion between source and observer. Formula: $f' = f \frac{v \pm v_o}{v \mp v_s}$ Used in radar & astronomy	FARADAY'S LAW Electromagnetic induction rule Induced EMF is proportional to the rate of change of magnetic flux. Formula: $\text{EMF} = -N \frac{d\Phi}{dt}$ Basis of generators & transformers	CENTRIPETAL FORCE Force toward circle's center The net force required to keep an object moving in a circular path. Formula: $F = mv^2/r$ Always directed inward	ESCAPE VELOCITY Speed to leave gravity Minimum speed needed for an object to escape a planet's gravitational pull. Formula: $v = \sqrt{2GM/R}$ Earth's value: ~11.2 km/s	BERNOULLI'S EQN Fluid flow energy conservation Describes conservation of energy in flowing fluid: pressure, kinetic, and potential. Formula: $P + \frac{1}{2}\rho v^2 + \rho gh = c$ Explains lift & flow speed	IDEAL GAS LAW Unifies gas behavior laws Relates pressure, volume, moles, and temperature of an ideal gas in one equation. Formula: $PV = nRT$ R is universal gas constant